SCRDR Sentiment Analysis

The Strategy of Grouped Keywords for NLP Generalization

The Challenge of Small Datasets

The Overfitting Problem

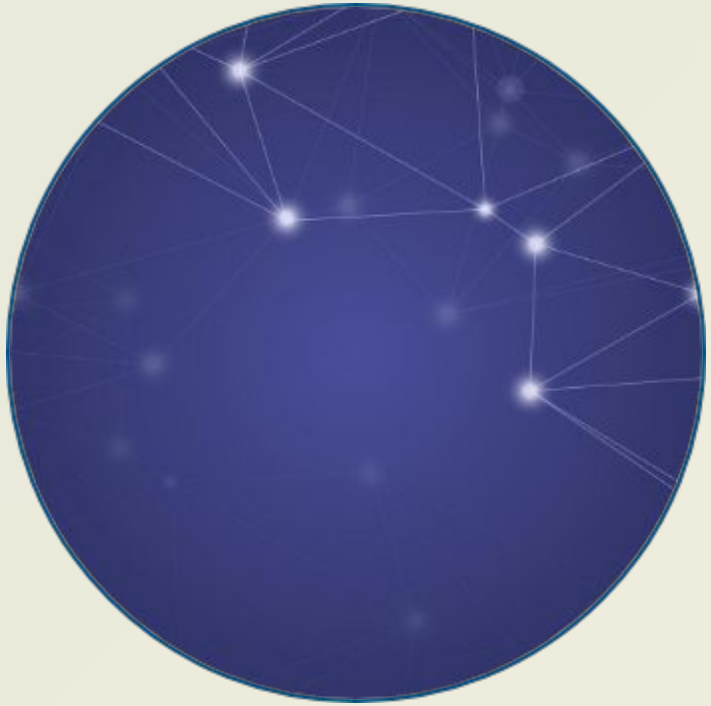
Sentence Memorization

Rules based on specific, long sentences lead to "Memorization" rather than learning. The model handles training data well but breaks on new input.

The Brittle Gap

Small variations in wording cause individual rules to fail. In low-resource settings, this results in extremely high complexity for low utility.

Regex Groups: The Solution



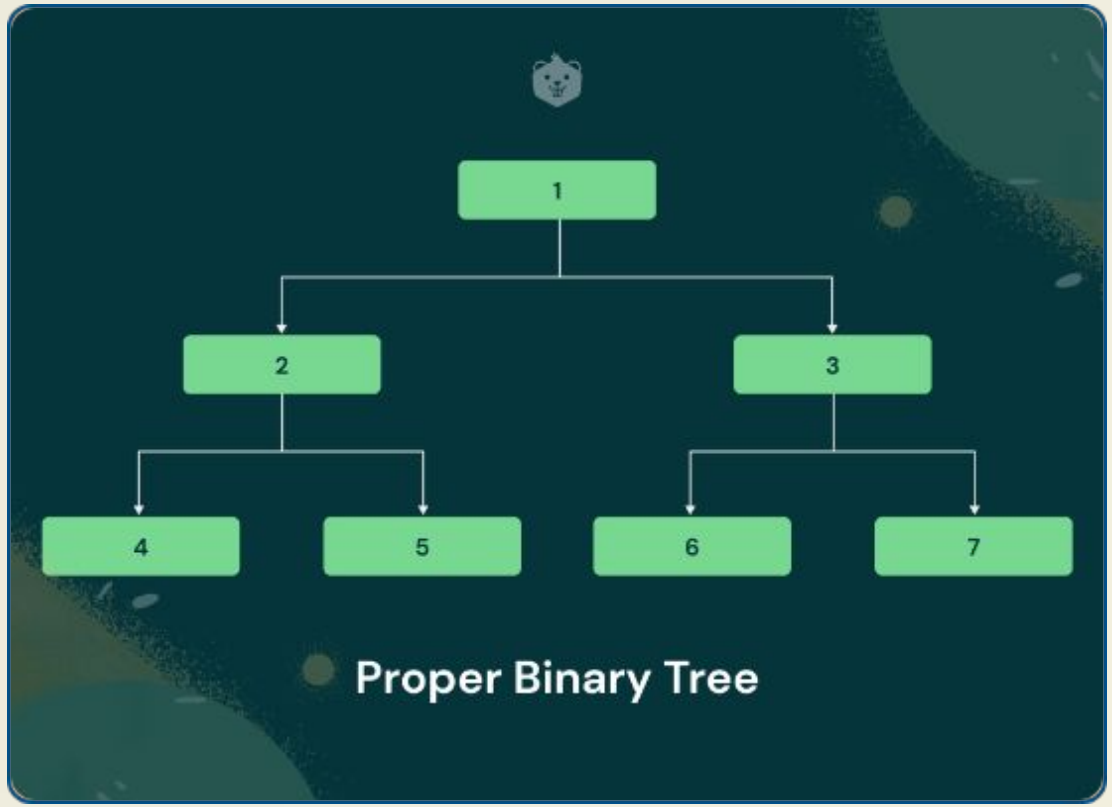
Mastering Generalization

By using the Regex OR operator |, we group synonyms and emotional markers into single, powerful rules. This mimics a dictionary-based approach within a rule-based SCRDR framework, allowing the model to recognize intent regardless of specific wording.

Flattening the Tree

Architectural Efficiency

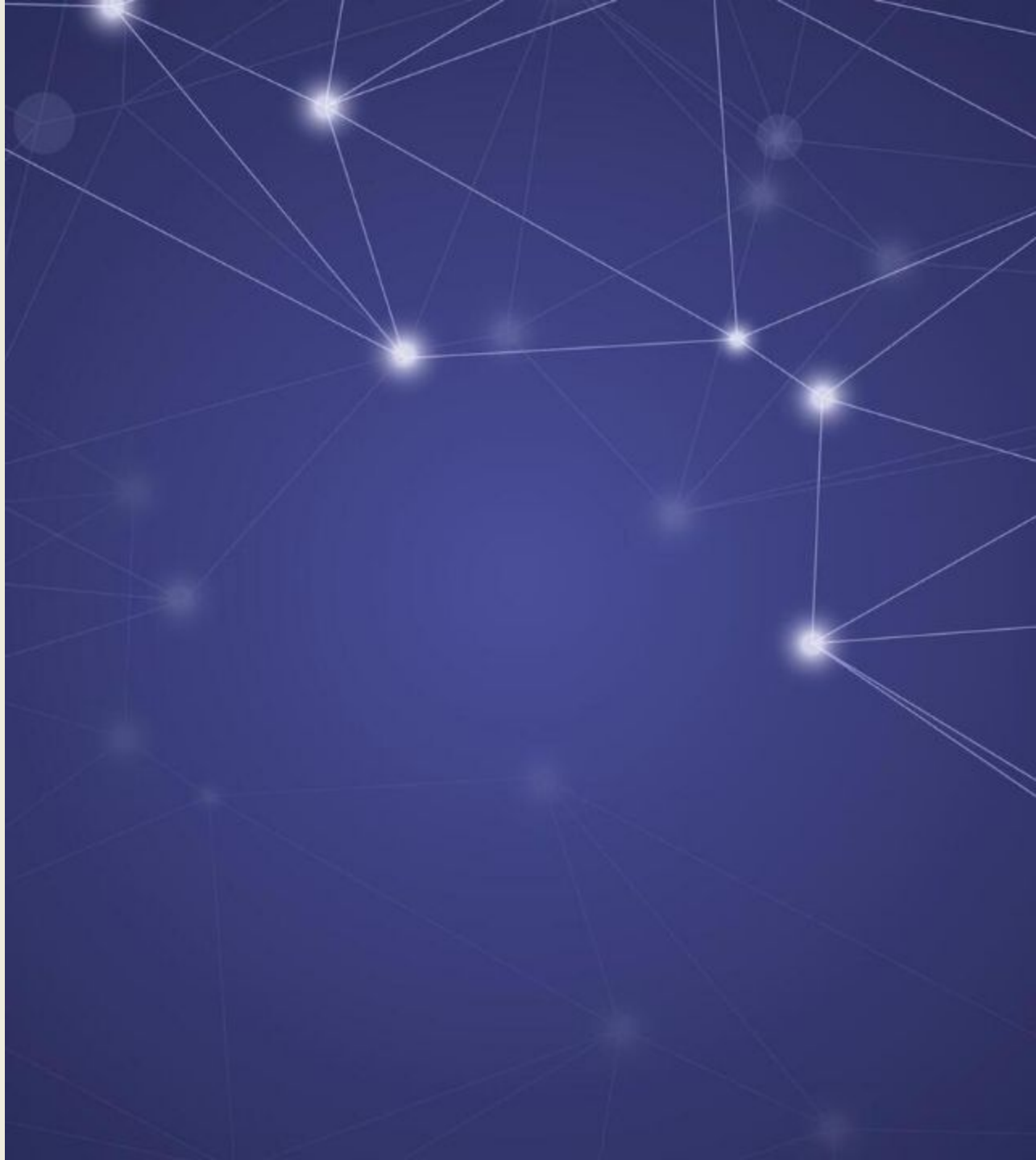
Grouped keywords create a "Flat Tree." Instead of an 80-node deep chain, we implement "Master Nodes" for primary emotions. This drastically reduces model complexity while improving maintenance and execution speed.



Linguistic Diversity

Myanmar language has diverse ways to express emotions. Sadness can be 'ဝမ်းနည်း', 'စိတ်ညစ်', or 'နာကျင်'.

Grouping these emotional roots ensures that the model recognizes emotional intent across various dialects and social media slang, bridging the gap between formal and colloquial text.



Training Performance

98%

Accuracy Achieved

Solid Foundations

The model achieved near-perfect performance on the training set, with balanced precision and recall across Anger, Fear, Joy, Love, and Sadness. Grouped rules proved immediately effective.

Perfect Test Generalization

The 100% test accuracy confirms that grouped keywords successfully learn general emotional triggers rather than memorizing phrases.

The Master Pattern Logic

Emotion	Grouped Regex Pattern (Key Emotional Markers)
Anger	(ဒေါသ|ယုတ်|လီး|စောက်|မအေ|သတောင်းစား|မကြိုက်)
Sadness	(နာကျင်|ငိုအဆင်မပြေ|ကွဲကြေ|အားငယ်|ဝမ်းနည်း)
Joy	(အဟီ|ဟားဟား|မိုက်|ဝမ်းသာ|ပျော်|ရင်တွေ.*ခုန်)
Fear	(ကြောက်|ထိတ်လန့်|စိုးရိမ်|ရင်ထိတ်|အေးစက်|လန့်)

Thank you for your attention.

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